# Unit Test Report

Online Exam Portal | Software Engineering Mini-Project, Part 3 | 10 November 2026

Student: Chiranthan V | SRN: PES2UG24CS138 | Section C

Release identifier: sprint2-final-2026-11-10 | Final submission report

Verified application commit: 14074716e3c2de2f59da3077ef265546004e704c

Final main CI revision: 14074716e3c2de2f59da3077ef265546004e704c. Backend, browser and performance runs below executed this merged revision.

The results below refer to these code versions. Earlier operational tests are listed separately with their dates.

## 1. Outcome and scope

PASS: 16 frontend component/function tests and 9 monitor regression tests. The supporting backend suite passed 101 tests on PostgreSQL, with no failures or skips. It includes API, integration and concurrency tests; these are not all isolated unit tests.

| Suite | Passed | Failed / skipped | Environment |
|---|---|---|---|
| Frontend Vitest | 16 | 0 / 0 | Node 24.22.2; Vitest 5.1.0; jsdom; 3 files |
| Monitor unittest | 9 | 0 / 0 | Python 3.11; clocks/mocks + local HTTP |
| Backend supporting regression | 101 | 0 / 0 | Python 3.11; PostgreSQL 16 CI |
| Earlier local backend cross-check | 95 | 0 / 5 | Windows Python 3.11.9; SQLite; original 100-test suite |
| Local frontend cross-check | 16 | 0 / 0 | Windows; existing installed Vitest 3.2.7 |

The earlier local run discovered 100 tests: 95 passed and five PostgreSQL-only cases were skipped. The recheck added a simultaneous-attempt-start test, so the current PostgreSQL run contains 101 passing tests, including six concurrency cases. The earlier local result is retained as a separate baseline, not relabelled as a run of the new test. Local monitor rerun: 9 passed in 3.721 seconds. PostgreSQL execution: 101 tests in 16.257 seconds. Results from different environments are not added together as unique cases.

## 2. Frontend unit/component results

| Test file | Tests | Result |
|---|---|---|
| api.test.js | 10 | passed |
| attempt-recovery.test.js | 3 | passed |
| forms.test.jsx | 3 | passed |

API tests check in-memory JWTs, CSRF/cookies, legacy attempt headers, concurrent refresh deduplication, session expiry, absent refresh cookie, partial-import errors, lockout, HTML error handling, network failure and pagination. Component tests cover distinct-option validation, stored text/server errors and correct-option selection. Recovery-age tests cover fresh/minute/hour values and unknown or reversed device-clock age.

## 3. Monitor regression results

All nine named cases passed: weighted complete-window accounting; overrun/missed slots; invalid URL/port/credentials/control characters; truncated response; TLS failure/socket timeout; concurrent summary-file nonoverwrite; invalid CLI/evidence reuse; invalid numeric configuration; actual local HTTP health/body/redirect behavior. No test disables TLS verification for real deployment evidence.

## 4. Supporting backend suite breakdown

| Test module | Passed | Purpose |
|---|---|---|
| test_api | 53 | Authentication, administration and attempt APIs |
| test_assessment_acceptance | 11 | Results, reports and proctoring acceptance |
| test_assessment_integration | 13 | Grading and attempt lifecycle integration |
| test_frontend_delivery | 3 | Frontend delivery and security headers |
| test_injection_security | 2 | SQL injection and stored-script regressions |
| test_part3_concurrency | 1 | Simultaneous starts and winning-session ownership |
| test_performance_tools | 2 | Benchmark statistics and seed safety |
| test_postgres | 5 | PostgreSQL row locks and concurrent requests |
| test_scheduled_notifications | 10 | Notification scheduling, retries and deduplication |
| test_sprint2_migration_compatibility | 1 | Preservation of existing attempts during migration |

The backend suite exercises authentication/lockout/RBAC/reset, roster imports, exam scheduling and selection, attempt binding and immutability, grading/reports/proctoring, notification outbox behavior, migration preservation and PostgreSQL races. Test dependencies are isolated fixture records and Django test databases; no real email receipt is implied.

## 5. Defects and limitations

Before release, dependency audit findings in the test/accessibility toolchain were patched in PR #67. Vitest 5.1.0 passes all 16 unit tests. Separate npm audits of both exact patched lockfiles reported zero known vulnerabilities at verification time. The audit JSON is retained; this is not a penetration-test or future-vulnerability guarantee.

The recheck found a gap in the test suite: duplicate-session behavior was checked sequentially, but simultaneous starts were not directly exercised. The new test uses two PostgreSQL transactions and a barrier inside the start path. It confirms one 201 response, one 409 response, exactly one attempt and start audit record, and resume access only for the winning session. The test passed; application code was not changed.

No assertion failures were found in the measured CI runs. The earlier confirmation-focus and sticky-timer defects were fixed in PR #60 and are covered by browser tests. Scoped team-lead acceptance is recorded separately from these automated results; no separate teammate or instructor sign-off is claimed. Local email tests check message generation; they do not establish delivery to a real mailbox.

## Evidence and reproduction

Team-lead acceptance record: https://github.com/chiranthanv/online-exam-portal/pull/61#pullrequestreview-5838992122 . PR #64 approval/review is recorded through chiranthanv; a separate Harkirat/Dwijesh approval is not asserted.

Backend CI: https://github.com/chiranthanv/online-exam-portal/actions/runs/42786113452

Frontend/browser CI: https://github.com/chiranthanv/online-exam-portal/actions/runs/42786113444

Repository evidence: docs/testing/part3/evidence/2026-11-10-release/. The manifest records SHA-256 hashes. Downloaded archives were checked against GitHub's artifact digests before extraction. The selected JSON and text results are preserved without report screenshots. The original 100-test run remains in evidence/2026-11-06/; the earlier 101-test recheck remains in evidence/2026-11-06-recheck/.

Detailed coverage: coverage.json, coverage.xml and backend-coverage-original.zip (includes the interactive htmlcov/index.html report). Backend log lists every test; frontend JSON lists each assertion/scenario and actual status.

```
python -m unittest discover -s scripts -p 'test_monitor*.py' -v
coverage run backend/manage.py test portal.tests --verbosity 2
coverage report
cd frontend
npm ci
npm test -- --reporter=default --reporter=json --outputFile=unit-results.json
```

Run backend commands from repository root with disposable PostgreSQL DATABASE_URL and DEBUG=1. Use requirements-dev.txt. The exact Windows interpreter used for local cross-checks was C:/Users/91984/Downloads/online-exam-portal/.submission-demo/runtime/python311/python.exe; the isolated backend path was prepended explicitly because that embedded runtime has a preconfigured search path.